

Calculated columns with DAX

DAX syntax best practices

Mastering DAX syntax

DAX syntax refers to the **rules** and **conventions** used to write DAX formulas. It includes **functions**, **operators**, and **expressions** that are used to manipulate and aggregate data.

01. Basics of DAX

The basic syntax of DAX is **similar to spreadsheet formulas**, with additional functions and operators specific to Power BI.

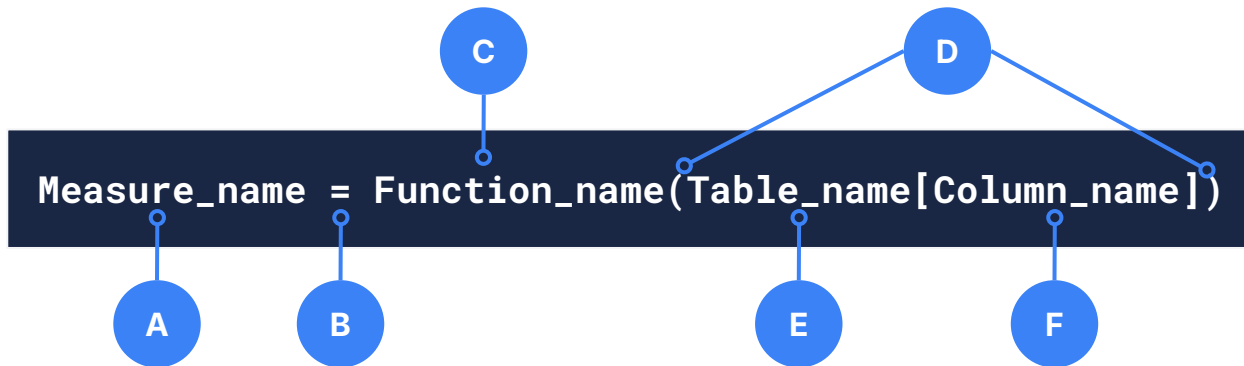
02. Why learn DAX syntax?

Learning how to create effective DAX formulas will help us **get the most out of our data** to solve real business problems that affect our bottom line.

03. Who should use DAX guidelines?

Whether a beginner or an experienced user, everyone using DAX should adhere to these guidelines to enhance the **clarity and robustness** of one's Power BI projects.

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A

The measure name.

B

The equal sign indicates the **beginning of the formula**. When calculated, it will **return a result**.

C

The **DAX function** to be used, for example, SUM.

Parentheses **surround an expression** that contains one or more arguments.

D

The referenced table.

E

The referenced column.

F

DAX syntax best practices

01. Use clear and descriptive naming

- When naming tables, columns, measures, or variables, choose names that **clearly convey the purpose or content**.
- Opt for a naming convention that **aligns with your project and team standards**.
- Always **avoid generic names or abbreviations** that may lead to confusion.



Do: clear and descriptive naming

```
Indicator_type =  
    RIGHT('Indicator'[Indicator_ID], 2)  
    == "EQ"
```



Don't: generic names and abbreviations

```
Column_7 =  
    RIGHT( 'Data_1'[IID], 2 ) == "EQ"
```

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02. Consistent naming conventions

- Consistency should **be applied across all elements**, including tables, columns, measures, and variables within your DAX formulas.
- Adopting a standardised approach to naming **ensures that anyone working** on the project can **quickly interpret and navigate** the codebase.
- This practice **promotes collaboration, reduces errors**, and **enhances the overall maintainability** of our Power BI solution.



Naming rules

- Within a single database all **tables must have unique names**.
- The names of the columns within each table must be unique:
 - Columns in different tables may possess the same names.
 - When the same column name is referenced from two or more tables, we must use a **fully qualified name** – that is, the 'Table_name' followed by the [Column_name].
- All objects (tables, columns, and measures) are **case insensitive**. For instance, the names 'DATE TABLE' and 'Date Table' would represent the same table.

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03. Format and indentation

- **Insert spaces around operators** to enhance code readability.
- **Use tabs to clearly indicate the hierarchy** of nested functions and improve code structure.
- When a function call has multiple arguments, **use carriage returns to separate** them. This improves the overall readability.



Do: readable

```
Combined_indicator_ID =  
    IF(  
        'Indicator'[Indicator_gender] == BLANK(),  
        'Indicator'[Indicator_ID],  
        "Male or female"  
    )
```



Don't: poor readability

```
Combined_indicator_ID=  
IF('Indicator'[Indicator_gender]==BLANK(),'Indicator'[Indicat  
or_ID],"Male or female")
```